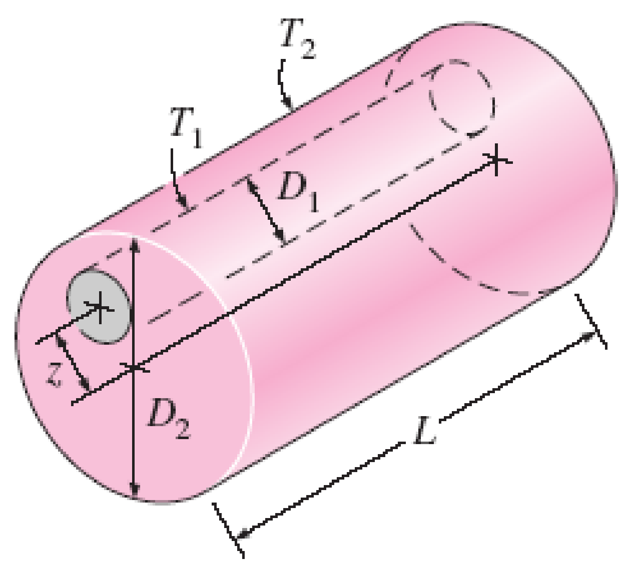
Flow inside eccentric annulus

Due 11th December 2023

Introduction

- Considerable interest has developed in the problems of flow and heat transfer in annuli, both concentric and eccentric.
- The interest in the eccentric annulus arises because of the problem of tube misalignment which frequently occurs in a close-packed tubular heat exchanger.



Introduction

- For laminar pressure driven flow, there exists analytic solutions.
- Snyder, W. T., & Goldstein, G. A. (1965). An analysis of fully developed laminar flow in an eccentric annulus. AIChE Journal, 11(3), 462–467. doi:10.1002/aic.690110319

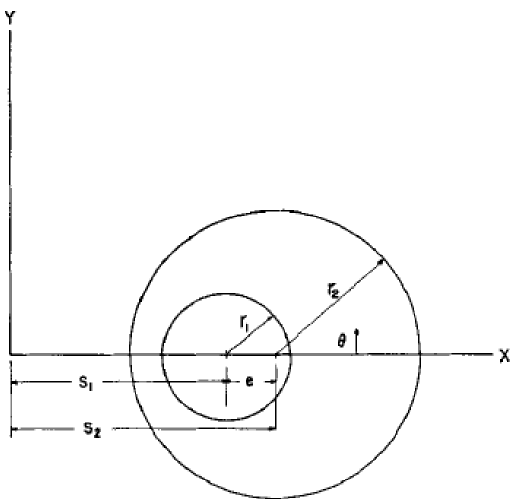
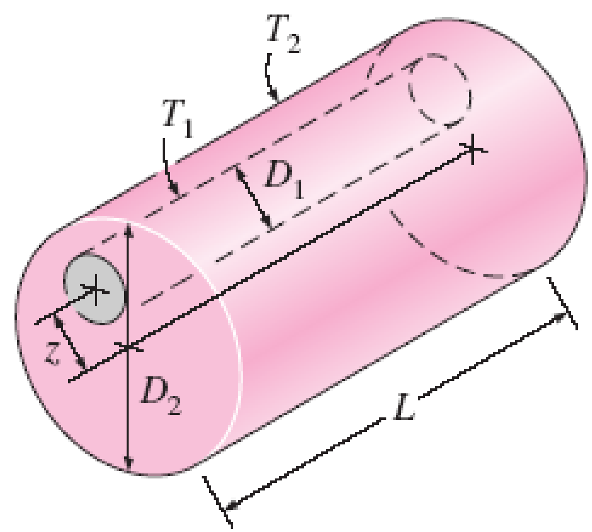


Fig. 1. Eccentric annulus geometry.



Introduction

- The bipolar coordinate is defined as,

$$x + iy = ic \cot\left(\frac{\xi + i\eta}{2}\right)$$

- The coordinate transformation

$$x = \frac{c \sinh \eta}{\cosh \eta - \cos \xi} \qquad (3a)$$

$$y = \frac{c \sin \xi}{\cosh \eta - \cos \xi} \qquad (3b)$$

$$e^{2\eta} = \frac{y^2 + (x + c)^2}{y^2 + (x - c)^2} \qquad (3c)$$

$$\tan \xi = \frac{2\, y\, c}{x^2 + y^2 - c^2} \qquad (3d)$$

$$y^2 + (x - c \coth \eta)^2 = \frac{c^2}{\sinh^2 \eta} \qquad (3e)$$

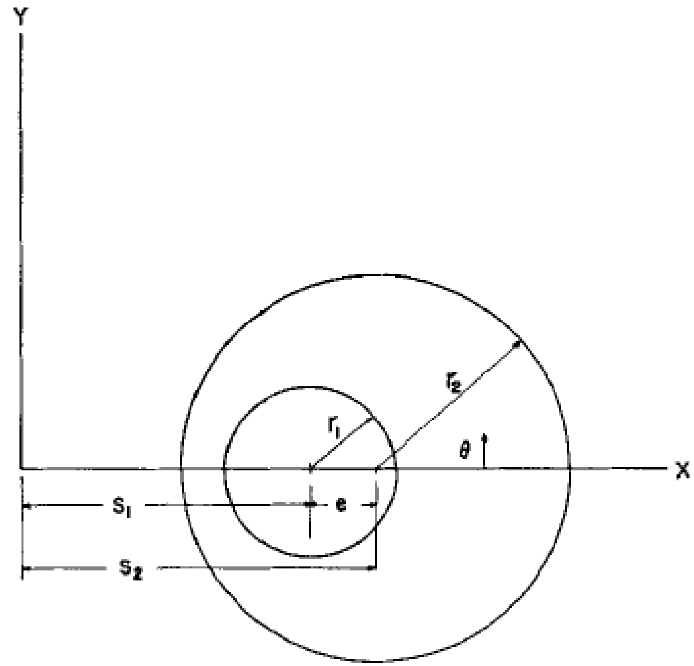


Fig. 1. Eccentric annulus geometry.

Equation (3*e*) shows that lines of constant η represent circles in the physical plane with center at $(c \coth \eta, 0)$ and radius $c/\sinh \eta$. The inner and outer surfaces of the annulus are thus represented by lines of constant η which will be designated as α and β, respectively. With this notation, it may be shown from geometrical considerations that the constants c, α, and β are given by the expressions

$$c = r_1 \sinh \alpha = r_2 \sinh \beta \tag{4a}$$

$$\cosh \alpha = \frac{1}{\gamma} \frac{\gamma(1+\phi^2) + (1-\phi^2)}{2\phi} \tag{4b}$$

$$\cosh \beta = \frac{\gamma(1-\phi^2) + (1+\phi^2)}{2\phi} \tag{4c}$$

$$\gamma = \frac{r_1}{r_2}$$

$$\phi = \frac{e}{r_2 - r_1}$$

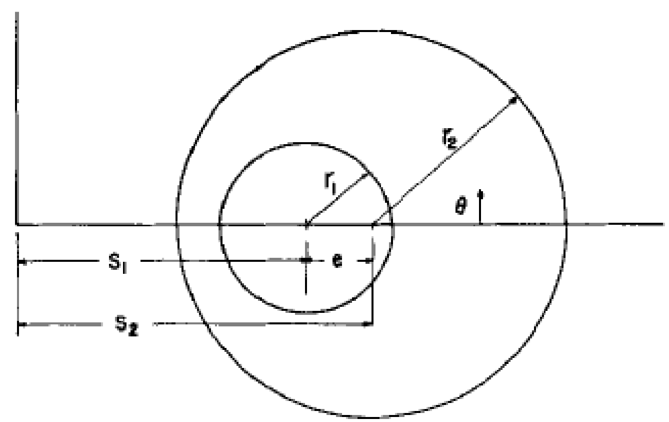


Fig. 1. Eccentric annulus geometry.

$v = -\dfrac{u}{\dfrac{c^2}{\mu}\dfrac{dP}{dz}}$ is the dimensionless velocity. The general solution to Equation (5a) is given in reference 7. Applying the boundary conditions $v(\eta = \alpha) = 0 = v(\eta = \beta)$ one obtains the general solution in the form

$$v = F + E\eta - \frac{ctnh\,\eta}{2} + \sum_{n=1}^{\infty} \{A_n e^{n\eta} + (B_n - ctnh\,\eta)\, e^{-n\eta}\} \cos n\,\xi \qquad (6a)$$

where

$$F = \frac{\alpha\, ctnh\,\beta - \beta\, ctnh\,\alpha}{2(\alpha - \beta)} \qquad (6b)$$

$$E = \frac{ctnh\alpha - ctnh\beta}{2(\alpha - \beta)} \qquad (6c)$$

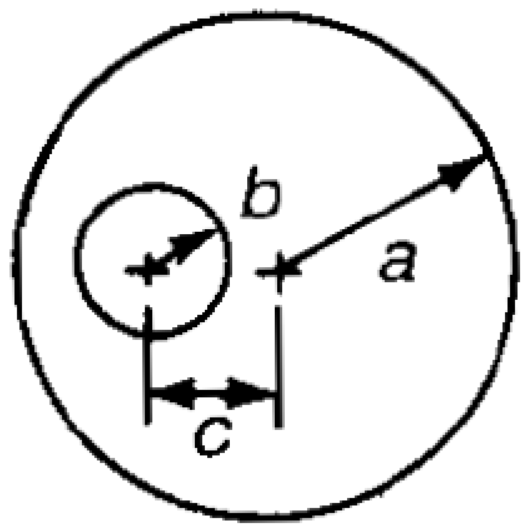
$$A_n = \frac{ctnh\alpha - ctnh\beta}{e^{2n\alpha} - e^{2n\beta}} \qquad (6d)$$

$$B_n = \frac{e^{2n\alpha}\, ctnh\beta - e^{2n\beta}\, ctnh\alpha}{e^{2n\alpha} - e^{2n\beta}} \qquad (6e)$$

For a given geometry, γ and ϕ, defined by Equations (4d) and (4e), would be specified. These values then determine α and β, given by Equations (4b) and (4c), respectively. With α and β determined, the constants appearing in Equation (6a), defined by Equations (6b) to (6e), are then fixed.

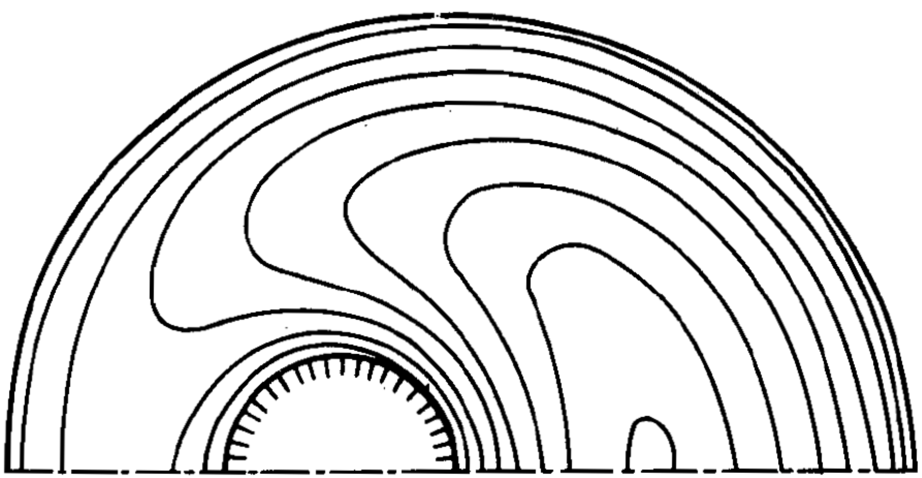
Problem to be solved by numerical simulations.

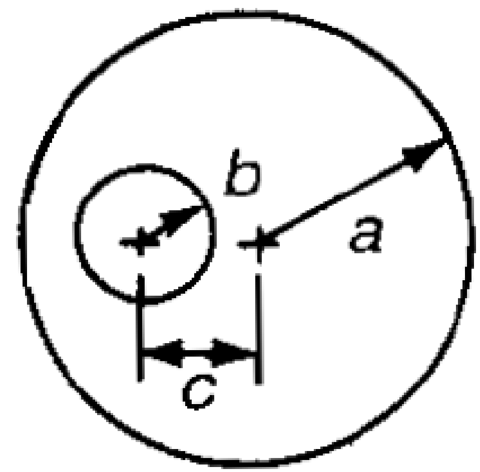
- The focus of the present task is to use the software ANSYS or others to simulate flow within laminar pressure driven eccentric annulus.



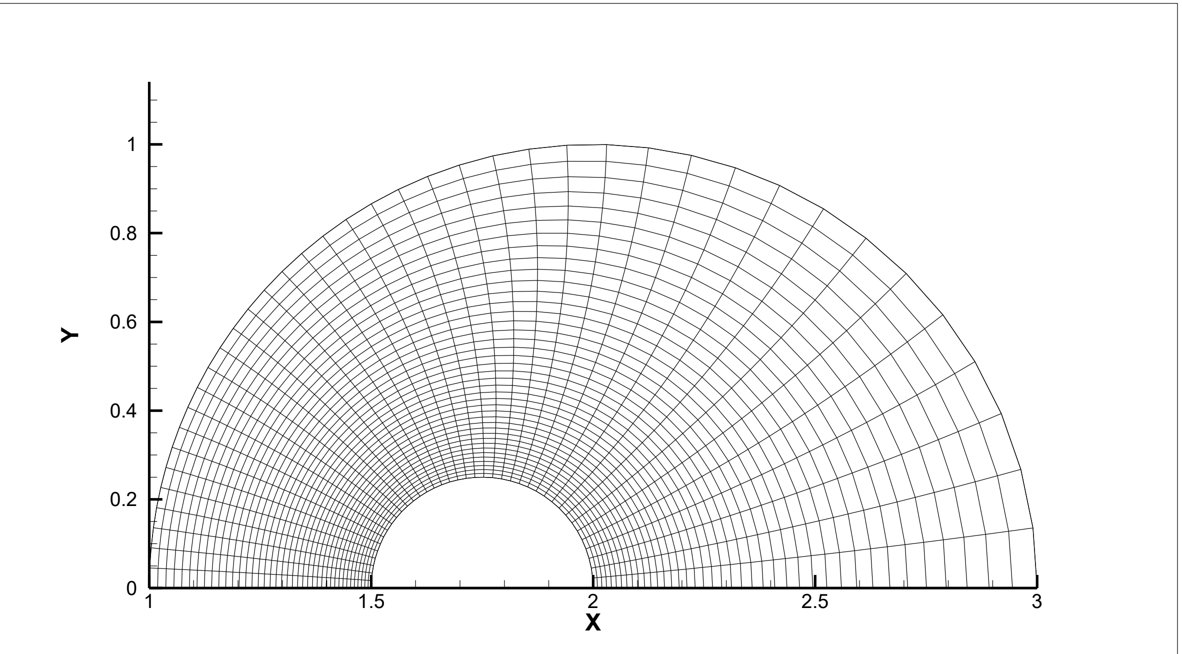
Problem to be solved by numerical simulations.

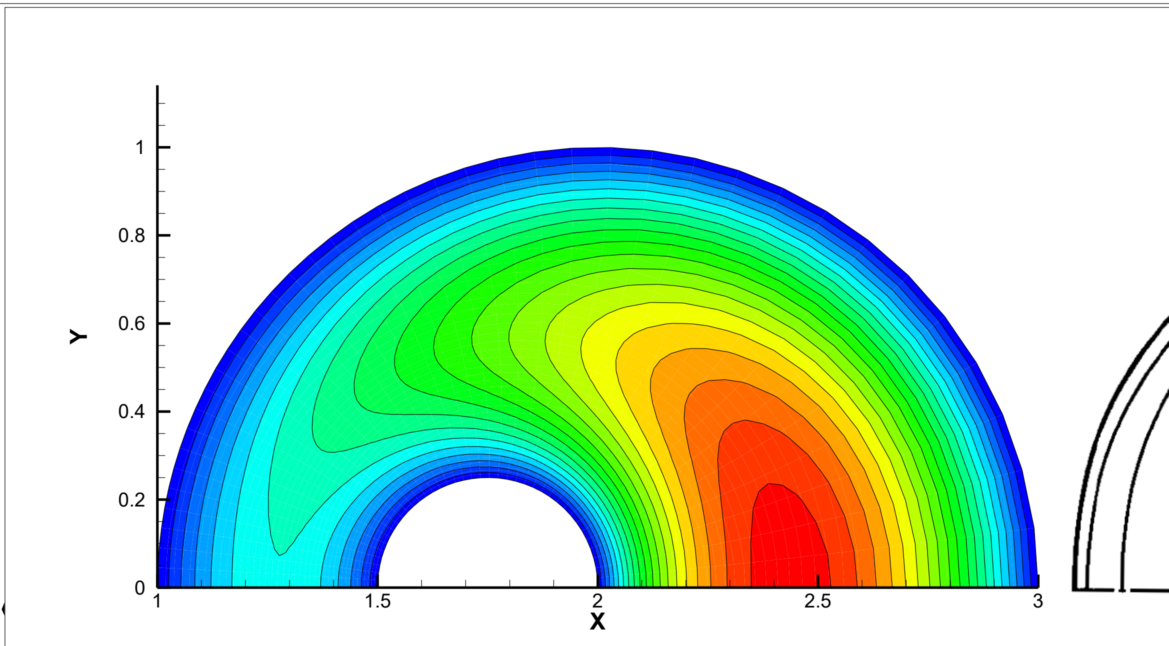
- You should compare your predicted results with the analytic solution.

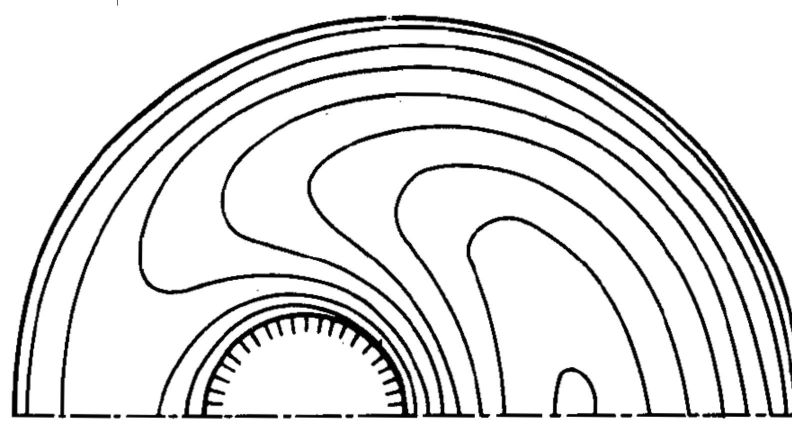




Constant-velocity lines for an eccentric annulus, $b/a = c/a = \frac{1}{4}$


Y
X
1
0.8
0.6
0.4
0.2
0
1
1.5
2
2.5
3


Y
X
1
0.8
0.6
0.4
0.2
0
1
1.5
2
2.5
3



Report format

- Introduction
- Brief methodology, such the software and schemes.
- Results and discussion
- Conclusions
- Please submit the report through the turnitin website.

How to use ANSYS

- PME has ANSYS in the computer lab at the 4^{th} floor of engineering building 1.
- If you have any problem regarding the usage of ANSYS, please consult the laminar flow TA at lab 606 or tel ext: 42603.
- You can also come to see me at my office.